

Algebra Exercises

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Chapter 8

Linear Algebra, Reprise

8.1 Functors

Exercise 8.1.1. Let $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ be a covariant functor, and let $\mathbf{C}$ and $\mathbf{D}$ have products. Show there exists a unique morphism $\mathcal{F}(A \times B) \rightarrow \mathcal{F}(A) \times \mathcal{F}(B)$ making the relevant diagram commute.

If $\mathbf{D}$ has coproducts (denoted $A \amalg B$) and $\mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ is contravariant, show there exists a unique morphism $\mathcal{G}(A) \amalg \mathcal{G}(B) \rightarrow \mathcal{G}(A \times B)$.

Solution. By definition of a functor, $\mathcal{F}(A \times B)$ is an object in $\mathbf{D}$, and

$$\begin{array}{ccc} & \mathcal{F}(A \times B) & \\ \mathcal{F}(\pi_A) \swarrow & & \searrow \mathcal{F}(\pi_B) \\ \mathcal{F}(A) & & \mathcal{F}(B) \end{array}$$

is a diagram in $\mathbf{D}$. Therefore, by universal property of products, the diagram

$$\begin{array}{ccccc} & & \mathcal{F}(\pi_A) & \xrightarrow{\quad} & \mathcal{F}(A) \\ & \nearrow & & & \uparrow \pi_{\mathcal{F}(A)} \\ \mathcal{F}(A \times B) & \dashrightarrow & \mathcal{F}(A) \times \mathcal{F}(B) & & \\ & \searrow & & & \downarrow \pi_{\mathcal{F}(B)} \\ & & \mathcal{F}(\pi_B) & \xrightarrow{\quad} & \mathcal{F}(B) \end{array}$$

commutes.

Now suppose $\mathcal{G} : \mathbf{C} \rightarrow \mathbf{D}$ is contravariant and $\mathbf{D}$ has coproducts. Since there exist projections $\pi_A : A \times B \rightarrow A$, $\pi_B : A \times B \rightarrow B$ in $\mathbf{C}$, and since $\mathcal{G}$ is contravariant, there exist morphisms

$$\mathcal{G}(\pi_A) : \mathcal{G}(A) \rightarrow \mathcal{G}(A \times B), \quad \mathcal{G}(\pi_B) : \mathcal{G}(B) \rightarrow \mathcal{G}(A \times B)$$

in $\mathbf{D}$. Since $\mathbf{D}$ has coproducts, by the universal property of coproducts,

$$\begin{array}{ccccc}
 \mathcal{G}(A) & & \xrightarrow{\mathcal{G}(\pi_A)} & & \\
 & \searrow i_A & & \nearrow & \\
 & \mathcal{G}(A) \amalg \mathcal{G}(B) & \dashrightarrow & \mathcal{G}(A \times B) & \\
 & \nearrow i_B & & \nwarrow & \\
 \mathcal{G}(B) & & \xrightarrow{\mathcal{G}(\pi_B)} & &
 \end{array}$$

exists in $\mathbf{D}$ and commutes. ■

Exercise 8.1.2. Let $\mathcal{F} : \mathbf{C} \rightarrow \mathbf{D}$ be a fully faithful functor. If A, B are in $\mathbf{C}$, show that $A \cong B$ if and only if $\mathcal{F}(A) \cong \mathcal{F}(B)$.

Solution. trivial, do later ■

Exercise 8.1.3. Formalize the notion of a presheaf of abelian groups on a topological space T . If $\mathcal{F}$ is a presheaf on T , elements of $\mathcal{F}(U)$ are called *sections* of $\mathcal{F}$ on U . The homomorphism $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ induced by an inclusion $V \subseteq U$ is called the *restriction map*.

Show that a presheaf is obtained by letting $\mathcal{C}(U)$ be the additive abelian group of complex valued functions on U , with restriction of sections defined by ordinary restriction of functions.

For this presheaf, prove that one can uniquely glue sections agreeing on overlapping sets. That is, if U, V are open sets and $s_U \in \mathcal{C}(U)$, $s_V \in \mathcal{C}(V)$ agree after restriction to $U \cap V$, prove that there exists a unique $s \in \mathcal{C}(U \cup V)$ such that s restricts to s_U on U and s_V on V .

Solution. Formalizing the notion is very simple. I will change notation slightly: let $(X, \mathcal{T})$ be a topological space, and define the category $\mathbf{T}$ such that $\text{Obj}(\mathbf{T}) = \mathcal{T}$ and

$$\text{Hom}_{\mathbf{T}}(U, V) = \begin{cases} \emptyset, & U \not\subseteq V \\ \{*\}, & U \subseteq V \end{cases}$$

Then a presheaf $\mathcal{F}$ on X is a contravariant functor $\mathcal{F} : \mathbf{T} \rightarrow \mathbf{Ab}$.

Let $\mathcal{C} : \mathbf{T} \rightarrow \mathbf{Ab}$ be the contravariant assignment given by

$$\mathcal{C}(U) = (C(U, \mathbb{C}), +).$$

That is, U maps to the set of continuous functions $f : U \rightarrow \mathbb{C}$, viewed as a group under addition. Morphisms $i : U \hookrightarrow V$ are defined as

$$\mathcal{C}(i)(f) = \rho_{VU}(f) = f|_U.$$

That is, the restriction to U of a section f of $\mathcal{C}$ over V is the standard function restriction. Verification that the group given is abelian is immediate, so now we show that the restriction morphisms are group homomorphisms. Note that if $f, g \in \mathcal{C}(V)$ and $U \subseteq V$, then

$$\rho_{VU}(f + g)(x) = (f + g)|_U(x) := f|_U(x) + g|_U(x) = \rho_{VU}(f)(x) + \rho_{VU}(g)(x).$$

Therefore, this is a group homomorphism by definition of the operation. Next, we show identity maps map to identity maps. Note that for any $f : U \rightarrow U$,

$$\rho_{UU}(f) = f_U = f.$$

Therefore, $\rho_{UU} = 1_{C(U)}$. Finally, we wish to show that if $U \subseteq V \subseteq W$, then $\rho_{WU} = \rho_{WV}\rho_{VU}$. Let $f : W \rightarrow \mathbb{C}$. We have

$$\rho_{WU}(f) = f_U, \quad \rho_{WV}\rho_{VU}(f) = \rho_{VU}f_V = f_U.$$

Therefore, $\mathcal{C}$ is a contravariant functor, and thus a presheaf of abelian groups on X .

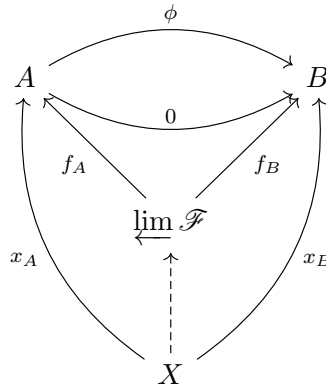
Finally, the gluing condition. Let $s_U \in \mathcal{C}(U)$ and $s_V \in \mathcal{C}(V)$ agree on $U \cap V$. We can define a function $s \in \mathcal{C}(U \cup V)$ as the map

$$s(x) = \begin{cases} s_U(x), & x \in U \\ s_V(x), & x \in V \end{cases}$$

Note that this is well-defined because if $x \in U$ and $x \in V$, then $s_U(x) = s_V(x)$. Moreover, it is manifestly unique since it is uniquely determined by s_U and s_V , and since they are continuous, s is clearly continuous as well due to the fact that s_U and s_V agree on $U \cap V$. Therefore, $\mathcal{C}$ is a sheaf of abelian groups on X . ■

Exercise 8.1.4. Verify that in $R\text{-Mod}$, the equalizer of $\phi : A \rightarrow B$ and $0 : A \rightarrow B$ is the kernel $\ker \phi$.

Solution. Verification is simple. Note that for all objects X , the diagram



commutes. Moreover, $f_B = 0f_A = \phi f_A$. However, since $0f_A = 0$, we have $\phi f_A = 0$, and $f_B = 0$.

We also have $x_B = 0x_A = \phi x_A$, and thus $x_B = 0$. We can rewrite this diagram as

$$\begin{array}{ccccc}
 & & 0 & & \\
 & \curvearrowright & & \curvearrowright & \\
 X & \xrightarrow{x_a} & A & \xrightarrow{\phi} & B \\
 & \searrow \text{dashed} & \uparrow f_A & \nearrow 0 & \\
 & & \varprojlim \mathcal{F} & &
 \end{array}$$

Note that this is precisely the universal property satisfied by the kernel, since by commutativity of the diagram, we have that $\phi f_A = 0$, and we can actually drop the 0 morphism $\varprojlim \mathcal{F} \rightarrow B$ while retaining precisely the same information. Therefore, the same proof given that kernels satisfy this universal property in $R\text{-Mod}$ suffices here to show that $\varprojlim \mathcal{F}$ exists, and is the kernel. ■

Exercise 8.1.5. Let $\mathbf{I}$ be the category consisting of elements of $\mathbb{Z}^+$ as objects, and a unique morphism $i \rightarrow j$ if and only if $i \leq j$. Let $\mathcal{F} : \mathbf{I} \rightarrow R\text{-Mod}$ be a functor, consisting of a choice of object A_i for each i , and a morphism $\phi_{ji} : A_i \rightarrow A_j$ for all i, j . Note that an element of

$$\prod_i A_i$$

is a sequence $(a_i)_{i>0}$. Call a sequence (a_i) coherent if $\phi_{i+1,i}(a_{i+1}) = a_i$ for all i . Let A be the submodule of $\prod_i A_i$ consisting of all coherent sequences. Let $\phi_i : A \rightarrow A_i$ be the restriction of the canonical projection $\pi_i : \prod_i A_i \rightarrow A_i$ to A . Show that A , together with the morphisms ϕ_i , satisfy

$$\varprojlim_i A_i = A.$$

Solution. First, I want to show A is truly an R -module. Let $(a_i), (b_i) \in A$. Then

$$(a_i) - (b_i) = (a_i - b_i).$$

Note that (with the mild abuse of notation), we have

$$\phi_{i+1,i}(a_{i+1} - b_{i+1}) = \phi_{i+1,i}(a_{i+1}) - \phi_{i+1,i}(b_{i+1}) = (a_i) - (b_i) = (a_i - b_i).$$

Additionally, for $r \in R$,

$$r(a_i) = (ra_i).$$

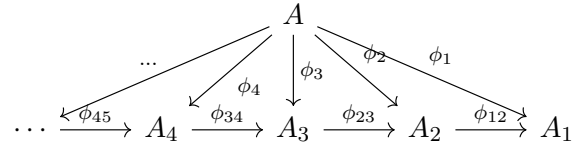
Thus,

$$\phi_{i+1,i}(ra_{i+1}) = r\phi_{i+1,i}(a_{i+1}) = r(a_i) = (ra_i).$$

Therefore, A is an R -module, as perscribed. Now we show the relevant diagram commutes. Let $n \in \mathbb{Z}^+$. We have $\phi_n(a_i) = a_n$. We also have

$$(\phi_{n+1,n}\phi_{n+1}) (a_i) = \phi_{n+1,n}(a_{n+1}) = a_n.$$

Therefore, the diagram



clearly commutes. Finally, A is manifestly final due to being the subset of a product and inheriting the canonical projections. ■